{EPITECH}

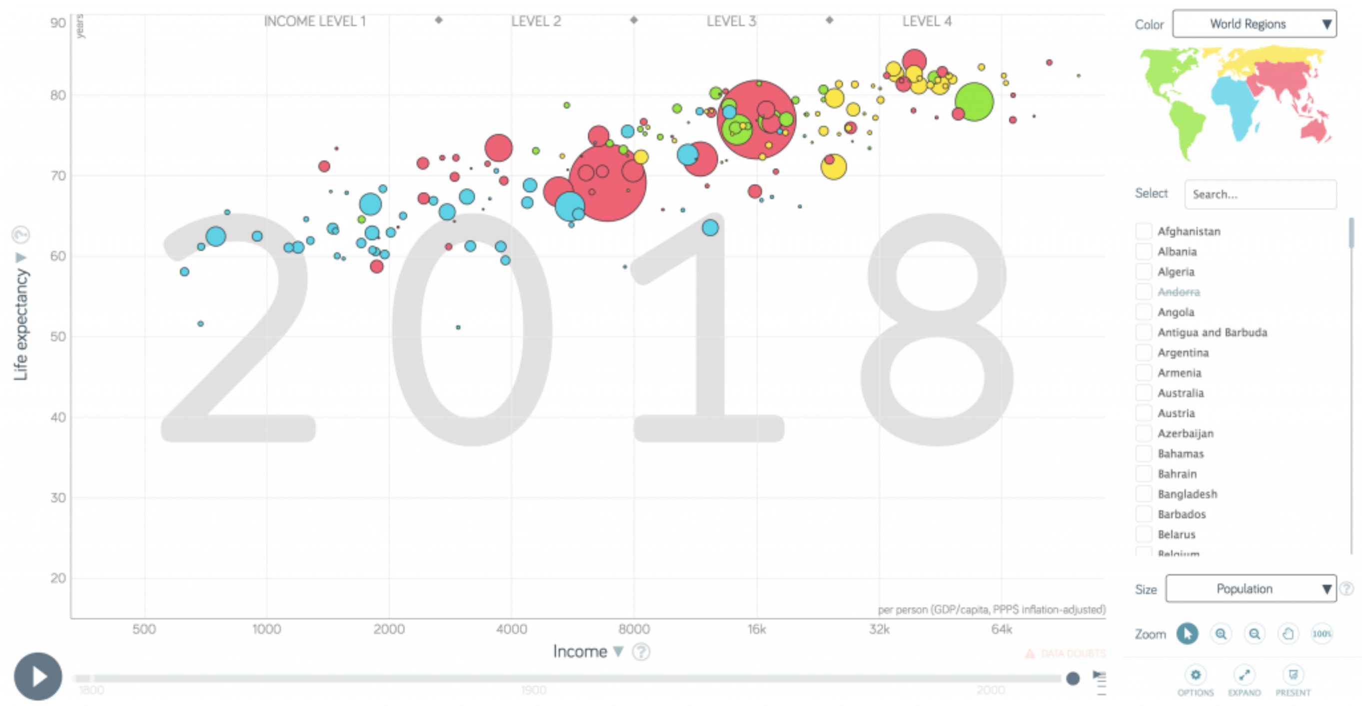
NEXTBUY_

< FROM RAW DATA TO SMART DECISIONS />

NEXTBUY

> The world cannot be understood without numbers.
> But the world cannot be understood with numbers alone.
>
> **— Hans ROSLING, Factfulness —**

Hans ROSLING was the co-founder and chairman of the Gapminder Foundation, which developed the fantastic and powerful Trendalyzer software system, also known as *Gapminder*.



Today, vast amounts of data are collected every minute.
But raw data is meaningless — *Muggles* can't make any sense of it.
As a data analyst, your job is to turn this gobbledygook into valuable insight and actionable strategies!

Test your skills on real-world data with millions of orders from thousands of grocery customers.
Each customer can place one or more orders. Each order contains one or more products.
Products are grouped into aisles. Aisles are grouped into departments.

Objectives & expectations

Investigate the datasets to **extract some actionable business insights** that drive revenue, marketing success, and operational efficiency.

You're **free** to decide what you want to dig, but we expect you to provide an Exploratory Data Analysis and build some predictive ML models.

You're provided with multiple datasets containing the following information:

Field	Description
add_to_cart_order	Product's order position in the cart (1 = first added)
aisle_id	Aisle's unique identifier
aisle	Aisle's name
days_since_prior_order	Number of days since user's previous order
department_id	Department's unique identifier
department	Department's name
order_dow	Order's day of week (0 = Sunday)
order_hour_of_day	Order's hour of the day (0-23)
order_id	Order's unique identifier
order_number	User's order sequence number (1 = first order)
product_id	Product's unique identifier
product_name	Product's name
reordered	Binary flag (1 = already ordered before, 0 = first time ordered)
user_id	User's unique identifier

To get started, you should:

- ✓ define at least 8 business insights you aim to provide, phrased as questions (see below);
- ✓ preprocess the provided datasets (combine, load, and clean them in a relevant order);
- ✓ analyze the data using statistics and charts to identify key products and shopping behaviors;
- ✓ build, train, test, evaluate and compare at least 2 predictive models.

To dig deeper, you could also:

- ✓ engineer new features;
- ✓ perform customers segmentation;
- ✓ implement pipelines to automate process;
- ✓ build a classification model for recommendations.

Here is a non-exhaustive list of questions to inspire you:

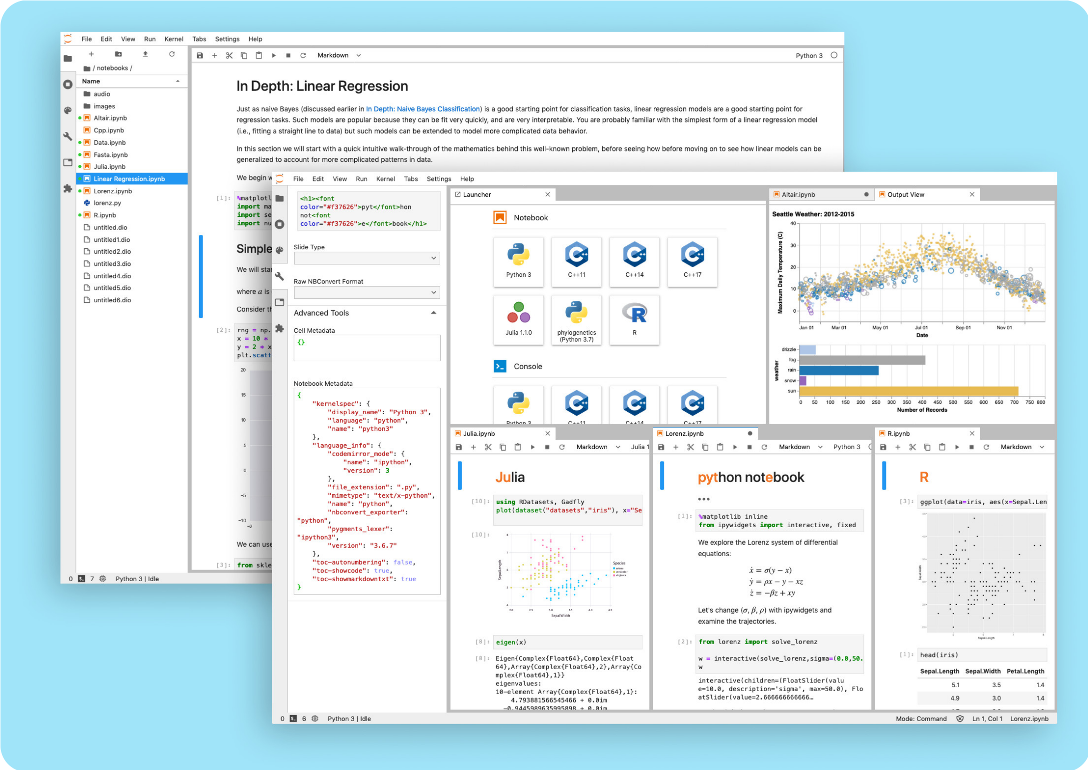
- ✓ Which products are the bestsellers?
- ✓ What is the proportion of organic food orders for vegetables?
- ✓ When and how do customers usually order?
- ✓ Can we predict whether a product will be reordered next time by a customer?
- ✓ Which products should be emphasized on Saturdays?
- ✓ How can we optimize in-store product placement to increase profits?
- ✓ What other products do users frequently order with chocolate?
- ✓ What are the main customer profiles, such as *midnight shopper*, *casual buyer*, or *cart addict*?
- ✓ What products should we recommend to a customer?
- ✓ Which item do customers put in their cart first?
- ✓ Which products have the highest probability of being reordered?
- ✓ Which aisle and department have the most (less) products?
- ✓ Are they pair of products or aisles that are highly correlated?
- ✓ Can we predict the average cart size of customer ordering bakery products?
- ✓ Do average customers order more cat food in the morning than dog food in the evening?
- ✓ Is there a relationship between time of last order and probability of reorder?

Imagine yourself as a sales manager for a product line and consider what insights they could leverage from your analysis and predictions.

Showing multiple times the same type of insights (such as wine bestsellers, meat bestseller, fruits bestsellers, ...) will ba count as one, nothing more.

Tooling

You MUST use Python and **Jupyter Notebook**.



> Some Integrated Development Environments have extension to manage Jupyter Notebook.

Except this imperative, you are free to use **any python packages you fancy**.

Be aware that someone will probably ask you to:

- ✓ **justify** why you choose a package rather than another one;
- ✓ **explain** briefly how your chosen packages work.

> **Your notebook must run on any machine, not just yours!**
> Make sure anyone can open it and run all cells successfully.
> Ask yourself: ”Could someone reproduce my environment from scratch without guessing?”
> Try to open and run your notebook on other machines.
> Optionally, convert your notebook into a more user-friendly, universally compatible format.

Delivery

Turn-in a `notebook.ipynb` containing your **E**xploratory **D**ata **A**nalysis of the given datasets, with code, visualizations, questions, answers, thoughts, remarks, etc... Your delivery must be as **organized** and **professional** as possible. So it must at least contain:

- ✓ some clear and concise explanations about your methodology and process for each step (problem definition, pre-processing, analysis, predictions, etc.);
- ✓ readable and understandable data visualizations, with their essentials components (such as axis, titles, legends,....) and without any chartjunk;
- ✓ guidelines to help anyone (even non-data scientists) quickly grasp the project.

> The interior decoration of graphics generates a lot of ink that does not tell the viewer anything new. The purpose of decoration varies [...]. Regardless of its cause, it is all non-data-ink or redundant data-ink, and it is often **chartjunk**.
>
> **— Edward TUFTE, The Visual Display of Quantitative Information - 1983 —**

Your notebook must present various types of dataviz. Each one is **accurate** and **appropriate**.
We do not simply use the same chart to compare, show distributions, and reveal relationships.

Bonus

Once you finished the main parts of this project, you can improve it in many ways, including:

✓ building interactive dashboards (Streamlit, Dash) to allow dynamic EDA and hypothesis testing;

✓ visualizing product relationships through interactive co-purchase networks and Sankey flows;

✓ using survival analysis to predict time until a customer's next order;

✓ creating advanced behavioral features such as basket similarity and time-between-orders statistics;

✓ training high-performance gradient boosting models with extensive feature engineering;

✓ constructing product-product graphs and extracting network features;

✓ applying anomaly detection methods to highlight unusual customer purchase patterns;

✓ building multi-level aggregations that combine user, product, and order history signals;

✓ exploring product or user embeddings with insightful multi-dimensional projections like t-SNE;

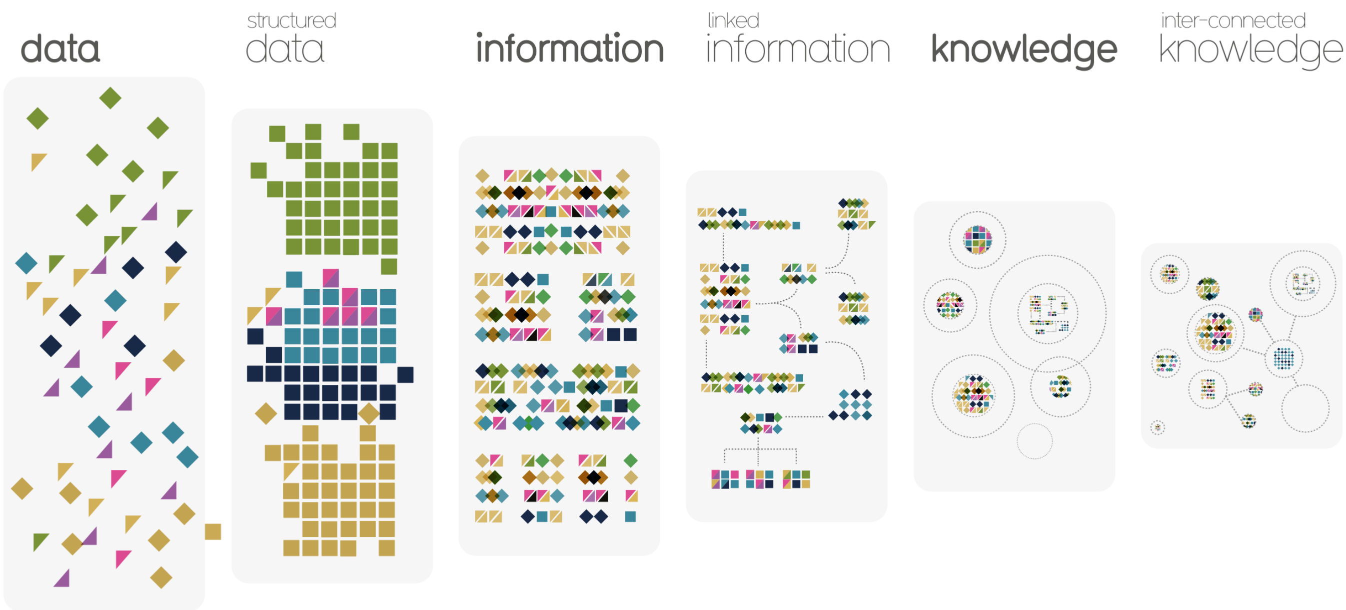
✓ leveraging deep sequential models (RNN, GRU, LSTM) to capture order histories over time;

✓ experimenting with model stacking or ensemble methods to squeeze out additional predictive power;

✓ automating hyperparameter optimization with tools like Optuna or AutoGluon;

✓ applying SHAP or permutation importance for deep model interpretability;

✓ performing Bayesian hierarchical modeling to capture variability across customers.



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